

Assignment 1

Understanding COLMAP's SfM Pipeline — Guided Exploration and Experiments

Objective: Explore a component of COLMAP's SfM pipeline in depth by answering a set of guiding questions about its role and design — by studying COLMAP's code — and back at least one answer with a hands-on experiment instead of an explanation alone.

Student Teams: Teams of up to 3. Sign up for a non-overlapping topic (first come, first served) via the shared sheet. Please be courteous to your classmates and not erase the preferences entered by others. Any conflicts will have to be resolved among the students.

Deliverables: 10-minute presentation per team, structured as **answers to your topic's guiding questions**, each grounded in specific code references. At least one answer must be supported by a **required experiment** — setup, results (plots/numbers), and interpretation.

Due Date: September 24, 2026

All topic references are with respect to the paper “*Structure-from-Motion Revisited*” (Johannes L. Schönberger, Jan-Michael Frahm).

1. Feature Matching at Scale

Match features across images for building the scene graph.

Reference: Section 2.1: Correspondence Search, Feature Extraction & Matching

Guiding questions to answer

- Why does exhaustive pairwise matching become infeasible at scale, and how does a vocabulary tree avoid it — what's actually being approximated?
- What does Lowe's ratio test protect against, and what happens to precision/recall as you tighten or loosen the threshold?
- Where does a vocabulary tree fail to retrieve a true match that exhaustive search would find? Can you construct such a case?

Experiment options

- Sweep Lowe’s ratio threshold on a real image set and plot precision/recall of matches (validated against geometric verification) at each threshold.
 - Construct a case where vocabulary-tree retrieval misses a true match that exhaustive search finds, and quantify how often this happens as database size grows.
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2. Geometric Verification & Scene Graph Augmentation

Validate matched image pairs and build a robust scene graph.

Reference: Section 2.1: Geometric Verification & Section 4.1: Scene Graph Augmentation

Guiding questions to answer

- What’s fundamentally different about what a homography H verifies versus what F/E verify — why does a valid H fit not imply the pair is safe to triangulate from?
- Why is a single model (say, just F) not enough to verify a pair — what failure mode does checking H alongside F and E catch?
- What do the N_H/N_F and N_E/N_F ratios actually indicate about the scene (e.g. planarity, pure rotation)?
- What is a “WTF” (watermark/degenerate) pair, and why can a naive verifier be fooled by one?
- With fixed thresholds, what kind of pair sits right at the decision boundary, and how does that ambiguity get resolved (or not)?

Experiment options

- Compute N_F , N_E , N_H across pairs spanning general, planar, and pure-rotation configurations, and show how the ratios classify them.
 - Find or construct a borderline pair near a threshold and show the classification flip as you perturb the threshold slightly.
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3. Initialization & Seed Pair Selection

Select the first image pair to start reconstruction.

Reference: Section 2.2: Initialization

Guiding questions to answer

- Why does a bad seed pair matter more than a bad pair selected later in the process?
- What's the tension between “many inliers” and “large baseline” in seed-pair scoring, and why do you need both?
- Can a pair with very high inlier count still be a poor seed choice? Construct or find an example.

Experiment options

- Score all candidate pairs in a dataset by COLMAP's criteria, identify its chosen seed, then run reconstruction from several alternative seed pairs and compare final reconstruction quality/completeness against the heuristic's choice.
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4. Next Best View Selection

Choose the next image to register for stable reconstruction.

Reference: Section 4.2: Next Best View Selection

Guiding questions to answer

- Why is “most matches” a bad greedy criterion for next-view selection on its own?
- What does the multi-resolution grid score capture that a flat 2D coverage count would miss?
- Does a registration order chosen by the heuristic ever get “stuck” (few strong candidates left) in a way an alternative selection strategy avoids?

Experiment options

- Compare final reconstruction completeness/accuracy under COLMAP's chosen registration order vs. a random order vs. a naive “most matches” heuristic, on the same dataset.
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5. PnP Solver & Pose Estimation

Register new images using 2D–3D correspondences.

Reference: Section 2.2: Image Registration

Guiding questions to answer

- Why does P3P need a minimal 3-point set, and what's the trade-off vs. using more points directly (e.g. EPnP)?
- What exactly is a cheirality check verifying, and what would go wrong without it?
- Why are near-planar 3D point configurations degenerate for PnP — what breaks mathematically?

Experiment options

- Inject increasing 2D–3D correspondence noise/outlier rate into a registration problem and plot pose error vs. noise level.
 - Construct a near-planar point configuration and show the PnP solver's behavior degrade/fail as planarity increases.
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6. Incremental Reconstruction & Pose Accumulation

Study the register → triangulate → refine → repeat loop that drives incremental SfM, and how errors propagate through it.

Reference: Section 2.2: Image Registration (incremental SfM overview)

Guiding questions to answer

- Why does incremental SfM register images one at a time instead of solving for all poses jointly from the start — what does it gain, and what does it risk?
- How does pose error for an image depend on when it was registered (early vs. late in the sequence)? Is drift monotonic, or does later BA correct earlier images?
- What triggers a local BA vs. a global BA within the loop, and why not just run global BA after every single image?
- Is there a point where accumulated drift makes a later image impossible to register at all, and how would you detect that happening?

Experiment options

- Snapshot camera poses after each new image is registered (`Mapper.snapshot_images_freq`) on a real capture, and plot how an early image's pose changes over the course of the reconstruction (before vs. after later global BA passes).
 - Vary `Mapper.ba_local_num_images` / `Mapper.ba_global_images_freq` and measure the effect on final pose accuracy and runtime.
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